

Design and Analysis of a Wireless Network Topology

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1. Introduction

This report presents the design and implementation of a wireless network topology created using a network simulation environment. The topology demonstrates how wireless networking devices such as Access Controllers (AC), Access Points (AP), wireless stations, and cloud connectivity can work together to provide wireless communication services.

The network is designed to simulate a real- world wireless infrastructure commonly used in universities, companies, and enterprise environments.

2. Objectives

The main objectives of this topology are:

- To design a wireless enterprise network.
- To understand the role of Access Controllers and Access Points.
- To provide wireless connectivity for client stations.
- To simulate centralized wireless management.
- To study communication between wireless devices.

3. Network Devices Used

Device Name	Device Type	Purpose
AC1	Access Controller (AC6605)	Centralized wireless management
AC2	Access Controller (AC6005)	Wireless control and configuration
AC3	Access Controller (AC6005)	Controls AP and wireless traffic
AP2	Access Point (AP2050)	Provides wireless access to clients
STA1	Wireless Station	Wireless client device
STA2	Wireless Station	Wireless client device
Cloud2	Cloud Device	Represents external network or internet
Cloud3	Cloud Device	Additional cloud connectivity simulation

4. Network Topology Description

The topology consists of multiple wireless network management devices connected together to form a centralized wireless infrastructure.

- AC2 is connected to AC1 .
- AC2 is connected to Cloud2.
- AC3 is connected to AP2.
- STA1 and STA2 communicate wirelessly through AP2.
- Cloud devices simulate internet or external network access.

This structure represents a managed wireless network environment where controllers supervise wireless access points and clients.

5. Wireless Communication

The wireless stations (STA1 and STA2) connect to the network using the Access Point AP2. The AP acts as a bridge between wireless clients and the wired infrastructure.

Wireless communication provides:

- Mobility for users.
- Flexible network access.
- Reduced cabling requirements.
- Easy scalability.

6. Access Controller Functionality

The Access Controllers (AC1 , AC2, and AC3) are responsible for centralized wireless management.

Their functions include:

- Managing wireless access points.
- Controlling wireless configurations.
- Monitoring wireless traffic.
- Enhancing wireless security.

- Supporting roaming and authentication.

Centralized management improves network performance and simplifies administration.

7. Access Point Role

The Access Point AP2 provides wireless connectivity to the client stations.

Main functions of the access point include:

- Broadcasting wireless signals.
- Allowing client association.
- Forwarding wireless traffic.
- Extending network coverage.

The AP acts as the communication bridge between wireless users and the wired network infrastructure.

8. Advantages of the Topology

This wireless topology provides several advantages:

- Centralized wireless management.
 - Easy monitoring and configuration.
 - Improved mobility for users.
 - Better scalability for enterprise environments.
 - Reduced infrastructure complexity.
 - Enhanced flexibility and accessibility.
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9. Practical Applications

This type of wireless network topology is commonly used in:

- Universities and schools.
- Hospitals.
- Office buildings.
- Enterprise campuses.

- Public wireless hotspots.
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10 . Conclusion

The implemented wireless topology successfully demonstrates how access controllers, access points, and wireless stations interact within an enterprise wireless environment.

The topology highlights the importance of centralized wireless management, mobility, and scalable wireless infrastructure. This practical implementation helps in understanding wireless networking concepts, enterprise WLAN deployment, and communication between wireless devices.